Digital Communications and Networks 7 (2021) 492–504

Contents lists available at ScienceDirect

Digital Communications and Networks

journal homepage: www.keaipublishing.com/dcan

Dynamic traffic congestion pricing and electric vehicle charging management system for the internet of vehicles in smart cities

Nyothiri Aung [a], Weidong Zhang [a], Kashif Sultan [b], Sahraoui Dhelim [c,*], Yibo Ai [a]

[a] National Center for Materials Service Safety, University of Science and Technology Beijing, Beijing, 100083, China
[b] Department of Software Engineering, Bahria University Islamabad Campus, Islamabad, Pakistan
[c] School of Computer and Communication Engineering, University of Science and Technology Beijing, Beijing, 100083, China

ARTICLE INFO

Keywords:
IoV
EV
VANET
Smart city
Congestion pricing
Congestion avoidance
EV charging
Traffic optimization

ABSTRACT

The integration of the Internet of Vehicles (IoV) in future smart cities could help solve many traffic-related challenges, such as reducing traffic congestion and traffic accidents. Various congestion pricing and electric vehicle charging policies have been introduced in recent years. Nonetheless, the majority of these schemes emphasize penalizing the vehicles that opt to take the congested roads or charge in the crowded charging station and do not reward the vehicles that cooperate with the traffic management system. In this paper, we propose a novel dynamic traffic congestion pricing and electric vehicle charging management system for the internet of vehicles in an urban smart city environment. The proposed system rewards the drivers that opt to take alternative congested-free ways and congested-free charging stations. We propose a token management system that serves as a virtual currency, where the vehicles earn these tokens if they take alternative non-congested ways and charging stations and use the tokens to pay for the charging fees. The proposed system is designed for Vehicular Ad-hoc Networks (VANETs) in the context of a smart city environment without the need to set up any expensive toll collection stations. Through large-scale traffic simulation in different smart city scenarios, it is proved that the system can reduce the traffic congestion and the total charging time at the charging stations.

1. Introduction

By the end of 2020, it is estimated that the U.S alone will lose $200 billion due to traffic congestion that was mainly caused by additional fuel consumption and travel time delay [1]. In addition to the huge economic losses, one of the main causes of CO_2 emissions and global warming is traffic-related congestion [2]. The emergence of Plug-in Hybrid Electric Vehicles (PHEVs) and Electric Vehicles (EVs) reduces fuel consumption to a certain degree. However, the random scheduling of EV charging in the charging stations may worsen the situation and cause traffic congestion as some of the charging stations are usually situated in the city center.

Generally, there are two types of traffic mitigation methods: infrastructure-based methods and rule-based methods [3]. Infrastructure-based methods require adding or modifying the roads network to redirect the traffic from the congested area to other roads. The rule-based methods reduce traffic congestion by regulating the usage time of the road network. There are several kinds of rule-based traffic reduction methods, including road rationing: banning certain kinds of vehicles from using the roads in some areas or at certain time intervals; parking restrictions: increasing the parking fees for vehicles that park in a congested area; congestion pricing: charging EVs for taking the congested roads during periods of the days like peak traffic hours. Congestion pricing is the most effective rule-based congestion mitigation method. There are two types of congestion pricing schemes, static scheme and dynamic scheme. The static scheme does not consider the time of road usage; therefore, the fees of using some roads are fixed all the time, which is not practical as it charges the driver for taking some empty road sometimes. In contrast, a dynamic scheme charges the drivers for taking some roads only during congestion times. Nonetheless, the major drawback of dynamic scheme is that these schemes can not motivate the drivers to take part in the congestion mitigation because the drivers are the main group of delays caused by traffic congestion, charging congestion toll to drivers will aggravate traffic congestion. In addition to the punishment mechanism, the system must also encourage the drivers simultaneously.

* Corresponding author.
E-mail addresses: nyothiriaung@yahoo.com (N. Aung), zwd@ustb.edu.cn (W. Zhang), kashif.buic@bahria.edu.pk (K. Sultan), sahraoui.dhelim@hotmail.com (S. Dhelim), ybai@ustb.edu.cn (Y. Ai).

https://doi.org/10.1016/j.dcan.2021.01.002
Received 12 November 2019; Received in revised form 20 January 2021; Accepted 25 January 2021
Available online 9 February 2021

Singapore is the first country that implemented a congestion pricing scheme in 1998, where a cash card uses Radio Frequency Identification (RFID) to detect and charge the vehicles that use the congested roads from 7:00 – 9:90 a.m. and 5:30 – 10:30 p.m. London city had also introduced a congestion pricing scheme in 2003, and the pricing scheme charges any private vehicle that takes a congestion road from 7:00 a.m. to 6:00 p.m. on weekdays. Stockholm city had also introduced a similar pricing scheme. All of these congested pricing schemes used old technologies like RFID to implement the control of the pricing system. On the other hand, the various applications of Vehicular Ad hoc Networks (VANETs) have opened the door for a new type of congestion pricing scheme that aligns with modern smart cities. VANETs offer an improved communication in the form of Vehicle-to-Infrastructure (V2I) and Vehicle-to-Vehicle (V2V) communications, and VANETs can gather and store traffic updates from-to EVs and RoadSide Units (RSUs) [4]. VANETs are used in many traffic-oriented smart city applications [5].

In this paper, we introduce a novel congestion-pricing scheme based on incentive-punishment rules, in which the drivers get rewarded by EV charging tokens for choosing to go through an alternative way to mitigate the traffic congestion. Furthermore, we introduce a token reward system that is used to reward the drivers for their contributions to reducing congestion. Tokens are also traded between drivers in the tender operations to manage the road and charging stations. The proposed method employs a dynamic pricing scheme to detect the traffic congestion in peak hours. By performing extensive simulations using a real map extracted from Beijing city, we have proved the effectiveness of the proposed system as a traffic congestion pricing and EV charging management system.

The contributions of our work can be summarized as follows:

1. We propose a congestion pricing scheme and an EV management system that both use road reservations. Therefore, the proposed scheme manages the roads using live traffic reports that are sent from all vehicles.
2. The proposed system uses reward-punishment rules to motivate the vehicles to go through alternative roads that could mitigate the traffic congestion near the electric charging stations.
3. The charging station reservation token can be exchanged between drivers using a tender operation that prioritizes urgent path requests.

The rest of the paper is organized as follows:

In Section 2, we present related works on congestion pricing and EV charging management. In Section 3, we present the main policies of the proposed scheme. In Section 4, we present the modeling design of the proposed scheme. Section 5 shows the performance evaluation as well as simulation details. In Section 6, the result of the simulation is discussed. Finally, in Section 7, we conclude the paper and present some of the future directions.

2. Related works

Many previous works have studied congestion pricing and EV charging management systems. In this section, we review the most relevant works that have studied the topic.

Atallah et al. [6] discussed the delay-optimal scheduling management of charging EVs at different Charging Stations (CSs) with several charging outlets. They proposed a centralized optimization model that is formulated based on the Integer Linear Problem (ILP) that accounts for the delayed arrival of EVs to charging stations and the randomness in the requested recharge interval time. Moghaddass et al. [7] introduced two scheduling schemes (static and dynamic) for optimal coordination of a group of cooperative EVs in a community with many CSs and potentially different chargers modes. The charging scheduling problems are modelled as mixed-integer multi-objective optimization models, and then multi-objective solution methods are used to find the optimal solution for each of the two scheduling schemes. Kabir et al. [8] discussed the photovoltaic (PV)-powered station that contains an Energy Storage System (ESS), which is supposed to be able of assigning variable charging rates to various EVs to fulfill their charging demands inside their expressed deadlines at minimum price. To ensure the fairness of pricing, a rate-dependent pricing scheme is proposed to ensure a higher price, which is conducive to a higher charging rate. The PV generation profile and future load requests are forecasted in each time slot to deal with their own uncertainties. An Integer Linear Programming (ILP)-based centralized system is studied to minimize the charging price per EV. Xie et al. [9] discussed the scenario of a car-sharing company that manages EVs and some parking lots. Clients can rent an EV in a given parking lot, then use it, and finally park it in another parking lot, and pay for the service at a certain price calculated by the company. The authors proposed a dedicated EV mobility scheme that can be used to monitor the spatial transportation of energy without tracking every single EV. Price elasticity is represented as a linear demand-price function. The company schedules the aggregated charging of free EVs in each parking lot to maximize the total profit. Milas et al. [10] introduced a framework of the smart charging of EVs, in which the smart charging uses a two-layer genetic algorithm that works in the CS with different charging modes that are connected to various charging locations in a resource-sharing manner. The advantages of the proposed methods are the fast optimization speed, the inclusion of user-defined factors, and feasible solutions that indicate the availability of the chargers in the CS. The communications between the EV and the CS are performed using the Open Platform Communications-Unified Architecture (OPC-UA) standard. Liu et al. [11] proposed a Distribution Locational Marginal Pricing (DLMP) method using chance-constrained Mixed-Integer Programming (MIP) designed to mitigate the possible congestion in the future distribution network with high penetration of EVs. To model the stochastic characteristics of the EV navigation patterns, a chance-constrained optimization of the EV charging is presented and formulated using MIP. Rigas et al. [12] studied the charging management of EVs at charging locations in a smart city to guarantee that the load at the charging location is still within the desired limits while minimizing the inconvenience to EV drivers. The authors develop solutions that consider charging locations and EV drivers as self-serving agents whose goal is to maximize their profits and minimize the impact on their schedules. Specifically, they propose variants of a decentralized and dynamic method and an optimal centralized static method. Huang et al. [13] leveraged genetic algorithms to find profit-maximizing station location and design details, with applications that reflect the costs of setup, management, and maintaining service devices, including land acquisition. Fast CSs are deployed across a congested city's network and follow stochastic demand for charging under a user-equilibrium traffic assignment. EV users' station choices consider endogenously determined travel times and on-site charging queues. The method allows for congested-travel and congested-station feedback into travelers' route selection under elastic demand and EV river' station selection, as well as the pricing elasticity of EV drivers. Luo et al. [14] introduced a CS location selection model to improve the resource usage of EVs for sustainable cities. In the proposed method, reservation services and idle rates during off-peak periods and waiting time during peak periods are considered. Finally, a used case from Chengdu, China, is used to illustrate the effectiveness of the proposed scheme. Li et al. [15] surveyed the related works on energy efficiency in femtocell networks, including energy efficiency metrics, deployments of femtocells, energy consumption models and energy-efficient schemes. In Ref. [16], they studied energy efficiency in cooperative cellular networks. Based on the double auction theory, they model the optimal relay assignment problem, which aims to enhance the performance of Cell-Edge Users (CEUs) with energy efficiency optimization. Finally, in Ref. [17], Resource Reservation and Allocation (RRA) with uncertain demands of mobile users is formulated as a robust optimization model. The logarithmic function is defined to obtain the mobile users' satisfaction, which shows how to match the allocations between RRs and VMRs according to the resource demands of the mobile applications. Aung et al. [18] proposed a dynamic congestion pricing scheme named T-Coin.

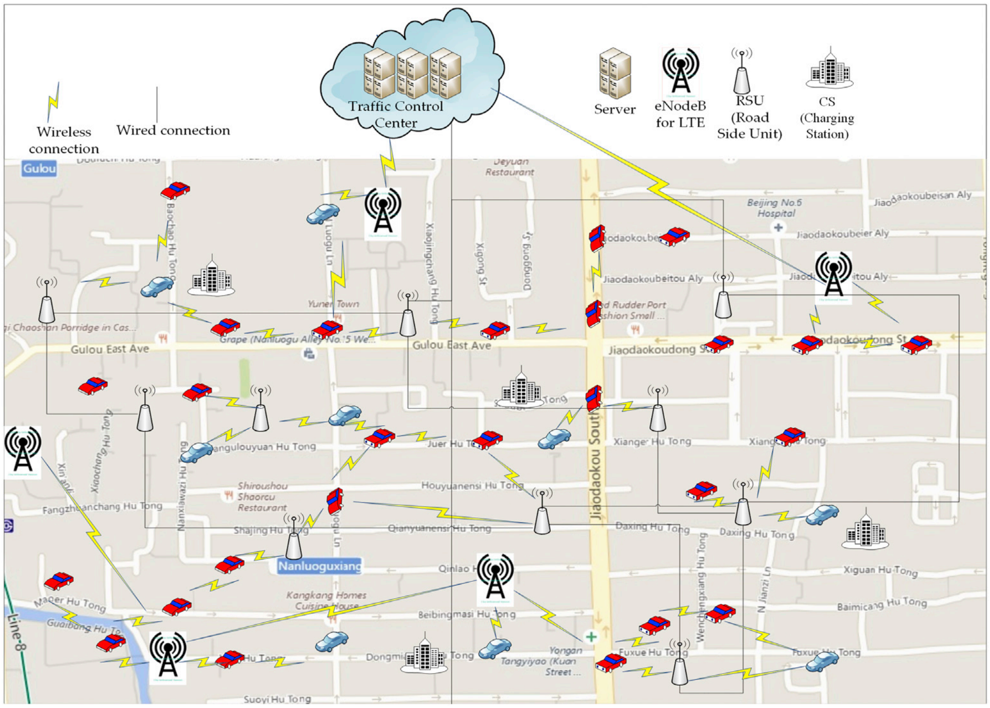

Fig. 1. Vehicular navigation architecture.

The main difference between the above-mentioned related works and the proposed system is that our proposed system uses a road reservation process along with the dynamic pricing to guarantee that the roads near the charging station are never congested, and also in our proposed system, the drivers have the choice to either take the relatively congested path and pay for using this road or use a relatively long path and earn some tokens that can be used for EV charging at CS.

3. System overview

In this section, we present a general overview of the proposed scheme.

3.1. Electric vehicular charging architecture

The specific infrastructure in the system under study and the VANET communication devices of the proposed scheme are presented in Fig. 1.

Traffic and EV charging management Center (TEVC): it is a centralized server that maintains the road reservation and EV charging requests. The TEVC manages and updates the Road Reservation and Charging Matrix (RRCM) that includes the details of the road reservations and the available positions in every road section and the charging request, in addition to the available charging slot at different time intervals. The TEVC updates the traffic information, including the average velocity per road section and the expected EV arrival time in the charging station. The TEVC also oversees the current road and charging slots demands, and estimates the current road pricing and maintains the token transactions between EVs in the case of the tender operation. For large-scale cities, a single TEVC may not be capable of maintaining the traffic updates and charging requests of such a large number of EVs. To tackle the scalability issue, the road network can be divided into multiple areas, where every area is controlled by its own TEVC, and the TEVCs communicate with one another to solve the cross-area requests. The architectural modeling of the TEVC is considered as future work.

Charging Station (CS): Charging stations are facilities that provide fast charging in a unidirectional Grid-to-Vehicle (G2V) power transfer, and it is also possible that the EV discharges its battery to the grid in exchange for tokens; this form of power transfer is known as Vehicle-to-Grid (V2G) power transfer. The CS is connected to the neighboring CSs and the TEVC. The latter is aware of all current and future EV positions, paths to destinations and charging demands. The main responsibility of the TEVC is to efficiently assign the EV to the optimal charging station by taking into account the traffic congestion on the path to the CS as well as the final destination.

RoadSide Unit (RSU): An RSU is a wireless communication gateway that links the EVs to the Internet. The RSU and the EV are connected using a Dedicated Short-Range Communication (DSRC). Generally, RSUs are placed near the intersection, and they are interconnected through a wired network, and RSU can be seen as the main communication network that links the wirelessly connected EV to the TEVC.

eNodeB: eNodeB is a micro base station that links the EVs to the 4G-LTE cellular network that makes it possible for the EV to access the TEVC in a ubiquitous manner. If the network is disconnected or one of the links of RSU-EV suffers from a communication failure as a result of DSRC range limitation, the EV can communicate with the TEVC using eNode.

EVs: To connect with RSUs and eNodeBs, all EVs are provided with a DSRC communication device integrated into their On-Board Unit (OBU). The EVs are also provided with a GPS-based navigation system that contains a digital roadmap of the smart city. The EVs inform the TEVC of their travel route changes in every road section. Since the route and location updates are extremely sensitive information, all the data exchanges between EVs and the TEVC are encrypted.

3.2. Token balance

To realize the proposed congestion pricing and EV charging management scheme, we propose to motivate the EV to cooperate with the system by offering them an incentive in the form of charging tokens that can be exchanged with charging service. The traffic tokens are also used as a punishment medium, as the proposed system also reduces the EVs' tokens in case they do not help the system to mitigate the traffic congestion. The traffic tokens are also used in the tender operations between EVs in the case of urgent charging requests. EVs can obtain traffic tokens using one of the following scenarios: (1) by directly purchasing

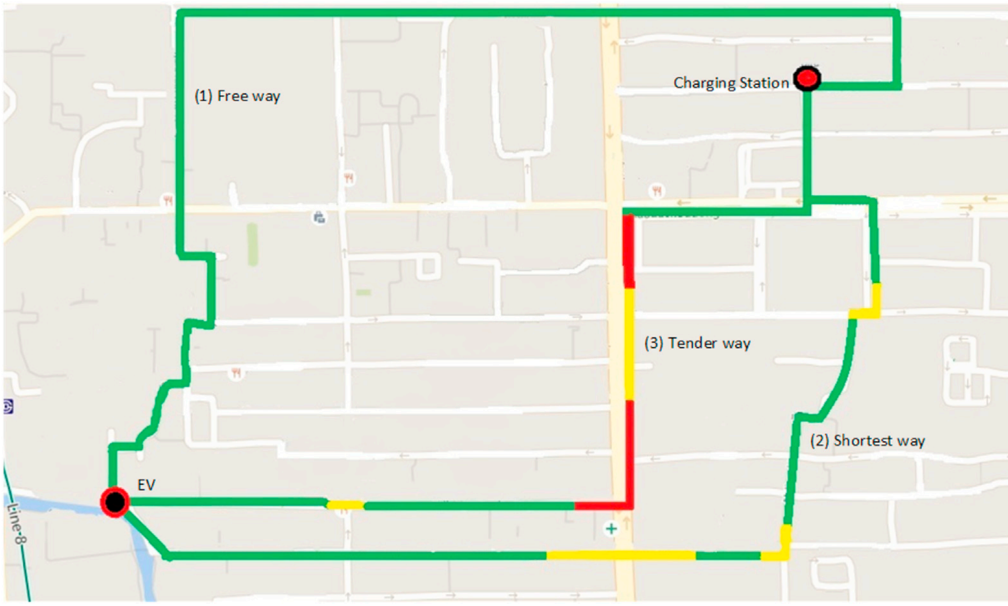

Fig. 2. The three types of EV traveling paths.

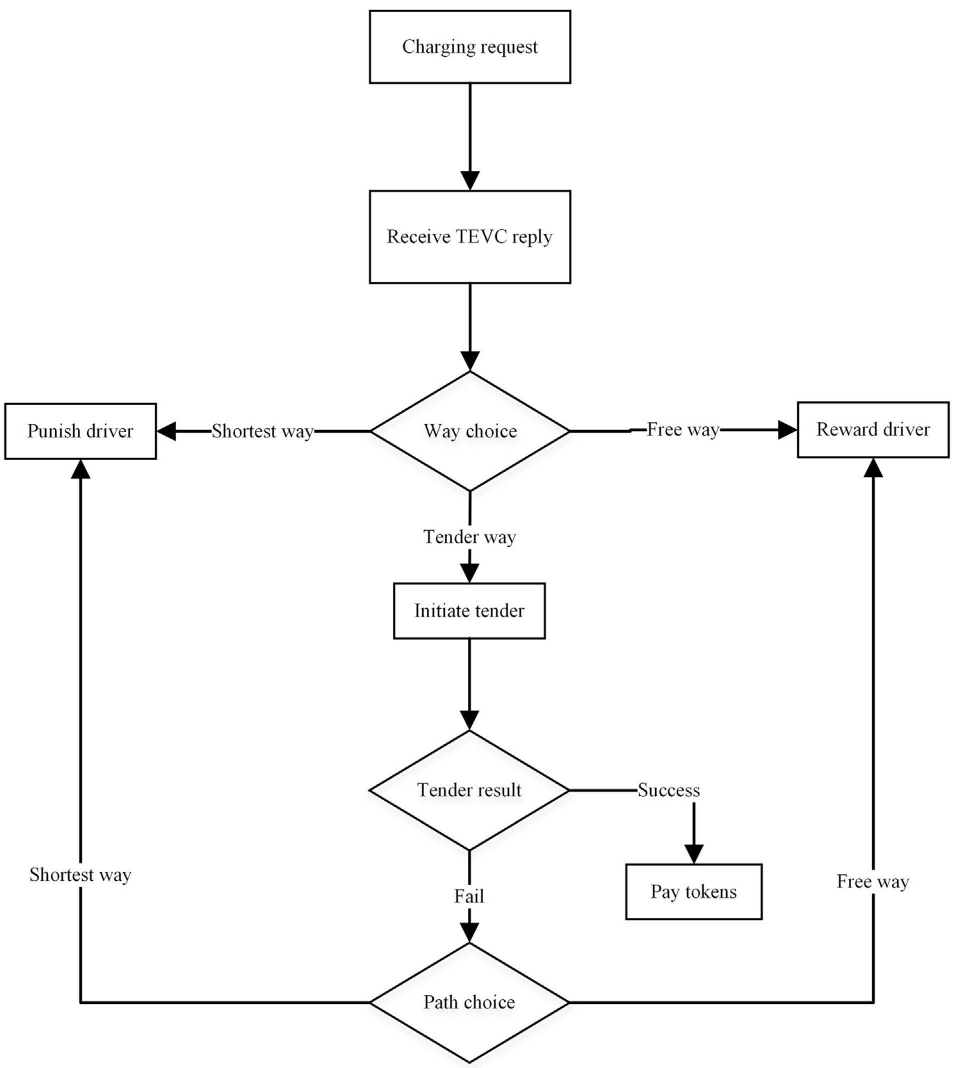

Fig. 3. The flowchart of path assignment.

with real money (which will also help the municipal authorities to generate more revenues), or (2) by acquiring it as an incentive for their contributions and good behavior, such as their willingness to go through another path that is not congested and relatively long, to mitigate the traffic congestion, or (3) in exchange for their road reservations or charging turn to other EVs that have urgent charging requests using a tender operation.

3.3. Charging reservation

To reduce the traffic congestion in the road near the CS, the capacity of the charging slot is divided into two parts. The first part of the charging slot quota can be reserved for free when the free quota is fully taken. After the free quota is completed, the EV that chooses to charge must pay a specific amount of tokens for charging. The charging slots and the roads are reserved by the first-come-first-served rule. That is to say, the roads and the CS are reserved for the first EV, which indicates that they want to pass or charge. When a new EV joins the network, it sends a charging request that includes the GPS locations of its departure and destination to the TEVC. The server calculates the shortest-way and the free-way (the shortest way only through the free traffic quota roads) from the source location to the destination point. The TEVC achieves this by computing the arrival time of each road section in both ways. Hence the TEVC can compute the traffic flow of every road section during different time intervals.

3.4. Incentive and punishment

After calculating the shortest-way ((2) in Fig. 2) and the free-way ((1) in Fig. 2), the EV will choose which way it will take to reach its CS destination. If the EV opts to go through the shortest way, it will be charged some tokens for taking congested path during peak traffic time and must pay the equivalent amount of charging tokens, whereas if the EV opts to go through the free-way, it will earn charging tokens equivalent to the time difference of travel delays between the shortest-way and the free-way (see Fig. 3).

3.5. Charging tender

In addition to the shortest-way and the free-way, the TEVC also suggests a set of road sections that were previously reserved. However, if they were not reserved, they would significantly reduce the travel delay from the source position to the destination CS. If the EV wants to pay extra tokens to go through these roads, the EV initiates traffic tender operation on these road sections, in such a case, the EV sends a message to all EVs in the current road section with the critical road sections' IDs to express its desire to pay a token to use the tender road section. The tender request message also contains the deadline of the tender operation. After receiving the tender request message offers, if the initiating EV confirms the offered tokens, it sends a way-update to the TEVC, and the confirmed bidders will get the specified tokens.

3.6. EV punishment

The proposed EV charging management scheme uses the road reservation to mitigate the congestion near the CS so as to ensure that road sections near the CS are used within their capacity limits. In order to take a road section, that road section must be within the pre-reserved way that was assigned to the EV by the TEVC. Any misuse of the road reservation operation will lead to more serious traffic congestions. The EVs that misuse the reservation rule and use the roads without prior reservations are punished by reducing their charging token balance, and even could suffer legal punishment by the smart city authorities if they repeatedly misused the system.

4. System design

In this section, we elaborate on the modeling design of the proposed scheme.

4.1. Map modeling

To format the city network road and prepare the calculation of the shortest way, the city map is represented in the form of a graph structure. Let $G = (V, E, C)$ be a directed weighted graph that denotes the road network of the smart city. Where E is the set of directed road sections, and V is the set of road intersections, C denotes the group of EVs in the city. The road sections are two-way, with one or more lanes on each side.

4.2. EV traffic flow

To design the model of driving delay and EV traffic flow, the charging time is represented as a group of time slots $T = \{t_1, t_2, \ldots, t_n\}$. The traffic situation on a specific road section is computed by calculating the traffic flow in that section during the designated time interval. The traffic flow in the road $r_{i,j}$ in the time slot t is denoted as $tr^t_{i,j}$ and is estimated by computing the mean of the speed $\gamma^t_{i,j}$ of all the EVs traveling in the traffic flow on this lane within the time interval t [19].

$$tr^t_{i,j} = E\left(\gamma^t_{i,j}(v_k)\right) \tag{1}$$

$\gamma^t_{i,j}(v_k)$ represents the speed of the EV v_k in the road $r_{i,j}$ during the time slot t. To compute the mean speed of traffic flow, the average speed of all EVs on the road $r_{i,j}$ is computed. In this context, we assume that the traffic conditions on a given road are consistent in each time slot. The information of the average velocity of every road lane is computed by measuring the EVs' traffic reports that they send to TEVC, as well as the EVs traffic computing using the roads' integrated loop detectors [20,21].

Let $d_{i,j}$ denote the delay time needed to go through the road section $r_{i,j}$, and $d_{i,j}$ is the ceiled number of time slots necessary for an EV to cross the road $r_{i,j}$ of length $l_{i,j}$ and the traffic situation $tr^t_{i,j}$, noting that $d_{i,j}$ also covers the waiting time at the intersection ω_j that the EV waits to pass the intersection j before passing to the following road $r_{j,k}$. Thus, the passing delay for the road $r_{i,j}$ is computed as shown in (2):

$$d_{i,j} = l_{i,j} \Big/ tr^t_{i,j} + \omega_j \tag{2}$$

Within a given time slot t_x, any road $r_{i,j}$ is assigned a weight, denoted as $w_{i,j}(x)$, that is updated dynamically at each time interval according to the road reservations, and it can be computed as shown in (3):

$$w_{i,j}(x) = C_{i,j} - \check{R}_{i,j}(x) \tag{3}$$

where $\check{R}_{i,j}(x)$ is the number of reserved positions in the road $r_{i,j}$ throughout the time interval t_x. And $C_{i,j}$ represents the total capacity of the road $r_{i,j}$, which is the maximum count of EVs that can go through road $r_{i,j}$ at the same time and with no traffic congested. Let $C_{i,j} = (c^1_{i,j}, c^2_{i,j}, \ldots, c^n_{i,j})$ be a set of EVs that are taking the road $r_{i,j}$ at the same time, we assume $r_{i,j}$ as a congested road if (4) is satisfied. In other words, if the difference between the total link delay time of all EVs in $C_{i,j}$ and the delay time of $r_{i,j}$ times by the number of EVs in the group n, is more than or equal to the road congestion threshold ε:

$$\sum k = 1 \;\; d^k_{i,j} - \left(n \times d_{i,j}\right) \;\geq \varepsilon \tag{4}$$

where $d^i_{i,j}$ is the lane delay time of vehicle c_i while crossing the road $r_{i,j}$.

4.3. Path reservation operation

4.3.1. Charging reservation matrix

Let R denote the group of all road sections in the smart city $R = \{r_0, r_1, r_2, \ldots, r_{N-1}\}$, and T is a sequence of consecutive time slots $T = \{t_0, t_1, t_2, \ldots, t_{M-1}\}$, the operation of reserving of R throughout T is managed by updating the road reservation matrix according to the road demand requests, the matrix is denoted as $M_{R,T} = (w_{r,t})_{m\times n}$, where $w_{r,t}$ is the weight of the road segment $r_{i,j}$ during time slot t. The weights associated with each road segment are updated according to the road demands and reservation operations, and $M_{R,T}$ is maintained and updated by the TEVC.

4.3.2. EV charging quota management

Similar to the road, the CS's charging position capacity is partitioned into two types of quotas (the free charging quota and the paid charging quota). The free charging quota can be reserved without the need to pay any tokens, and when the free charging quota is completely taken, EVs that opt to use the CS need to pay a defined amount of tokens to use the CS. The capacity of the paid charging quota changes dynamically based on the charging demand at the previous time intervals. The more the charging demand increases, the more its paid charging quota grows. To model the charging quotas dynamic management, we have proposed a charging quota allocation algorithm. The proposed algorithm is adapted from the communication protocol TCP's congestion control algorithm named Additive Increase Multiplicative Decrease (AIMD), and our proposed algorithm is shown in Algorithm 1.

Algorithm 1. Charging Quota Allocation

Algorithm 1 Charging Quota Allocation

```
1: σ = ⌈C_{i,j}/100⌉
2: if (M_{r_{i,j},t} < Q^T_{i,j}) then
3:    δ ← (M_{r_{i,j},t} − Q^T_{i,j})
4:    Q^c_{i,j}(t+1) ← Q^c_{i,j}(t) + δ
5: else
6:    if (Q^c_{i,j}(t) < Q^T_{i,j}) then
7:       Q^c_{i,j}(t+1) ← Q^c_{i,j}(t) − σ
8:    end if
9: end if
```

Whereas $Q^T_{i,j}$ is the paid charging quota threshold, it is a defined number that is to note the limitations of the paid charging quota when the charging demand is more than this threshold. The paid charging quota for the current CS will increase dynamically based on the upcoming charging demand. $Q^c_{i,j}$ is the currently paid charging quota. σ denotes the additive growth parameter, and we have set it as 2% of the total charging capacity of the CS. And δ is the multiplicative reduction parameter, which we set to the number of EVs using the paid charging quota.

4.4. Dynamic pricing

4.4.1. Punishment pricing

The most important question that one may ask is "what is the token count (price) of taking a given road section or using a given CS".

The price in terms of tokens is not fixed, and it changes dynamically based on pricing strategy. In our proposed scheme's pricing strategy, the amount of token payment depends mainly on 3 factors: (1) the length of the road section the EV will take, (2) the current road demand during the reservation request time interval, and (3) the significance of the current lane in the EV's path. The function of dynamic pricing is defined as shown in (5):

$$\begin{aligned}\phi^v_{i,j}(t) &= f(l_{i,j}) \times g\left(Q^p_{i,j}(t)\right) \times h\left(r^v_{i,j}(t)\right)\\ &= \tilde{l}_{i,j} \times \tilde{d}_{i,j}(t) \times \tilde{r}^v_{i,j}(t)\end{aligned} \tag{5}$$

where $f(l_{i,j})$ is the pricing function that correlates the road length to the corresponding amount of charging tokens,

$$\begin{aligned}&f(l_{i,j}) \to \tilde{l}_{i,j}\\ &\tilde{l}_{i,j} = l_{i,j} \times \tau\end{aligned} \tag{6}$$

τ is the toll rate, and is computed by the smart city authorities. In our simulation, we supposed that $\tau = 1 Token/km$.

$g(Q^p_{i,j}(t))$ is the function of road demand, and it is the linear transformation of the charging demand on the road section $r_{i,j}$ during time interval t to the road demand factor, which is denoted as $\tilde{d}_{i,j}(t)$.

$$\begin{aligned}&g\left(Q^p_{i,j}(t)\right) \to \tilde{d}_{i,j}(t)\\ &1 < \tilde{d}_{i,j}(t) < 2\\ &\tilde{d}_{i,j}(t) = \frac{\left(c_{i,j} - Q^p_{i,j}(t)\right)}{c_{i,j}} + 1\end{aligned} \tag{7}$$

$h(r^v_{i,j}(t))$ is road significance function, and it denotes the significance of road $r_{i,j}$ in the EV v's path to its destination, h calculates the optimum alternative way from intersection i to v's destination location, represented as $\ddot{p}^r_{i,s}(t)$. The output of h is the significance factor, represented as $\tilde{r}^v_{i,j}(t)$, and it is calculated according to the delay ratio of $\ddot{p}^r_{i,s}(t)$ and $p_{i,s}(t)$ as shown in (8). That is to say, if the delay of the alternative way is more than two times of the delay if EV v goes through the road $r_{i,j}$, then this road is seen as a critical road for EV v, and the significance factor will be 1. Hence it won't increase the total cost $\phi^v_{i,j}(t)$.

$$\begin{aligned}&h\left(r^v_{i,j}(t)\right) \to \tilde{r}^v_{i,j}(t)\\ &1 < \tilde{r}^v_{i,j}(t) < 2\\ &\tilde{r}^v_{i,j}(t) = \begin{cases} 1, & \dfrac{d\left(\ddot{p}^r_{i,s}(t)\right)}{d\left(p_{i,s}(t)\right)} > 2\\[2ex] \dfrac{d\left(\ddot{p}^r_{i,s}(t)\right)}{d\left(p_{i,s}(t)\right)}, & \text{else}\end{cases}\end{aligned} \tag{8}$$

4.4.2. Incentive pricing

The incentive for opting to avoid the congested way is represented as $\delta^v_{i,j}(t)$ and is calculated as shown in (9):

$$\delta^v_{i,j}(t) = \begin{cases} w_{i,j}(t) > Q^p_{i,j}(t), \ \delta^v_{i,j}(t) = \dfrac{f(l_{i,j}) \times g\left(Q^p_{i,j}(t)\right)}{h\left(r^v_{i,j}(t)\right)}\\[2ex] \text{else}, \ \delta^v_{i,j}(t) = 0\end{cases} \tag{9}$$

5. System evaluation

In this section, we show the evaluation baselines, evaluation metrics, as well as the simulation environment.

5.1. Baselines

To measure the effectiveness of the proposed congested pricing and EV charging management scheme, we have compared the performances of the following baselines:

Dijkstra Fixed Tokens (DFT): Since Dijkstra shortest way algorithm is the most famous path assignment method. We have designed a scheme

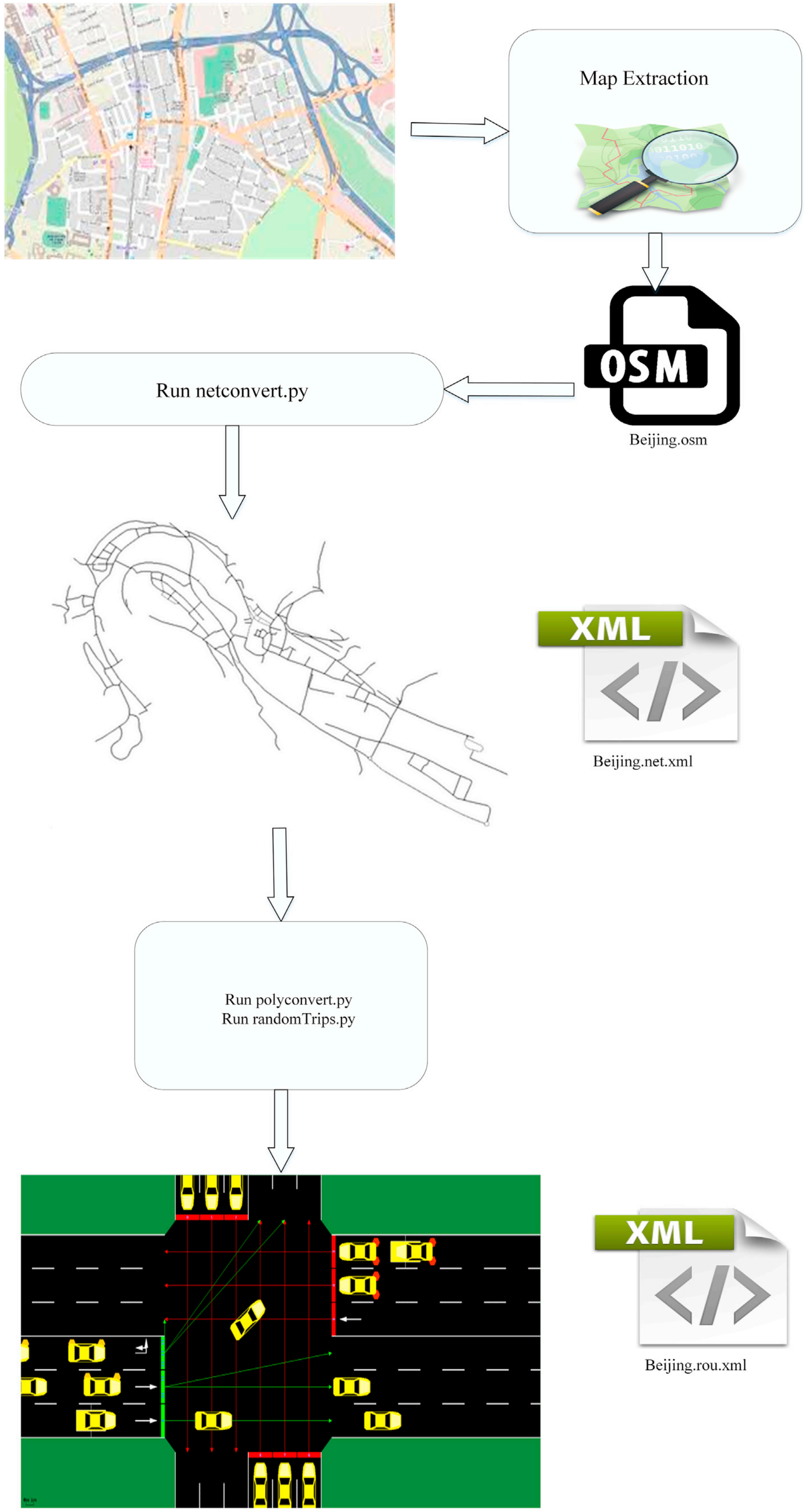

Fig. 4. EVs trip management using OSM and SUMO.

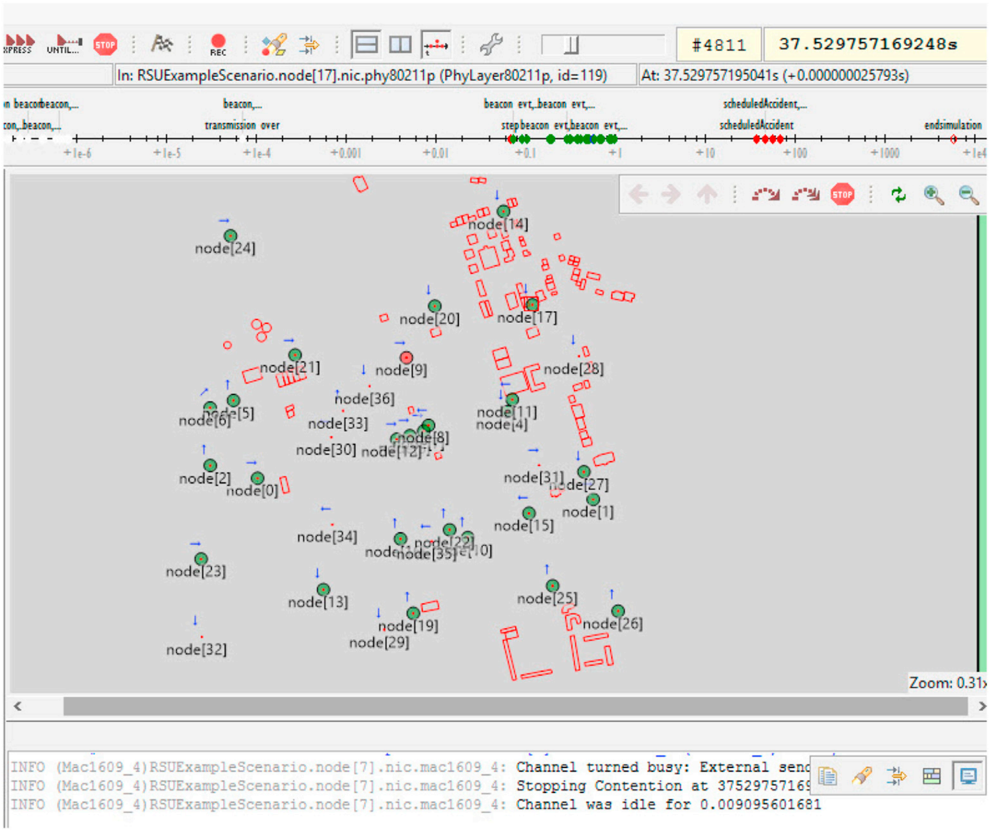

Fig. 5. OMNET++ simulation using Veins framework and SUMO.

(from now on denoted as DFT) that is based on the Dijkstra algorithm to compute the shortest way from starting location to the destination; the EV will take that way regardless of the traffic congestion within the roads of the way and near the CS where every road section has a fixed quota.

Fixed Pricing Tokens (FPT): To measure the effectiveness of adaptive pricing and charging quota management method of the proposed scheme, we have designed a static version of the proposed scheme (from now on denoted as FPT) and included it as a comparison baseline, in which we make the paid charging quota as a fixed amount, hence the paid charging quota is still the same regardless of the congestion.

Adaptive Pricing Token (APT): This is the proposed scheme (from now on denoted as APT).

5.2. Evaluation metrics

We have included the following comparison metrics to test and compare the effectiveness of the above-mentioned baseline methods.

Additional distance (Traveled distance vs Shortest distance): This metric is meant to compute the additional traveled distance that the EV has taken compared to the distance of its shortest way from the starting location to the destination. That is, we calculate the percentage of the actually taken distance from the starting position to the target position compared to the distance from the starting location to the destination location using the shortest way. We represent the additional distance of the EV v_i as $AD(v_i)$, as seen in (10), and the total additional distance of all EVs is computed as shown in (11).

$$AD(v) = \frac{\sum_{i=0}^{m-1} l_i^f}{\sum_{i=0}^{n-1} l_i^s} \tag{10}$$

where $p_i^s = \{r_1^s, r_2^s, \ldots, r_n^s\}$ and $p_i^f = \{r_1^f, r_2^f, \ldots, r_m^f\}$ are the shortest way and the shortest free-way of the EV v_i respectively, l_i^f and l_i^s are the lengths of road r_i^f and road r_i^s, respectively.

$$AD(V) = \frac{\sum_{i=0}^{p-1} AD(v_i)}{p} \tag{11}$$

where $V = \{v_0, v_1, \ldots, v_{p-1}\}$ is the group of all simulated EVs.

Additional time (Traveling time vs Estimated navigation time): This metric is to compute the additional time that the EV has used in order to reach the CS compared to the preliminary estimated time of the shortest way. That is to say, we calculate the percentage of the actual traveling time from the starting location to the destination location, and the traveling time starting location to the destination using the shortest way. We represent the additional time of EV v_i as $AT(v_i)$, and it can be calculated as shown in (12), and the total additional time of all EVs is calculated as shown in (13).

$$AT(v) = \frac{\sum_{i=0}^{m-1} d_i^f}{\sum_{i=0}^{n-1} d_i^s} \tag{12}$$

where $p_i^s = \{r_1^s, r_2^s, \ldots, r_n^s\}$ and $p_i^f = \{r_1^f, r_2^f, \ldots, r_m^f\}$ are the shortest way and the shortest free-way of the EV v_i respectively, and d_i^f is the travel delay of road section r_i^f, and d_i^s is the travel delay of road section r_i^s.

$$AT(V) = \frac{\sum_{i=0}^{p-1} AT(v_i)}{p} \tag{13}$$

where $V = \{v_0, v_1, \ldots, v_{p-1}\}$ is the group of all EVs in the smart city.

Token gain: This metric is to compute the total earned T-Coin, in which we calculate the total earned tokens from the EV punishments and the total lost tokens that are given as an incentive for mitigating the congestion. The token gain is computed as shown in (14).

$$G = \sum_{r=0} P_r - \sum_{r=0} R_r \tag{14}$$

Table 1
Simulation parameters.

Parameters	Description
Network Simulator	Omnet++5
Traffic Simulator	Sumo 0.27.1
Map Information	OSM
Simulated Location	Beijing
Simulated Area	5 km^2

Table 2
EV wireless communication parameter.

Parameter	Value
PHY model	802.11p
Channel frequency	5.890e9 Hz
Propagation model	Two ray
MAC model	EDCA
Propagation distance	450 m
Maximum hop	15
Fading model	Jakes model rayleigh fading
Shadowing model	LogNormal
Antenna model	Omnidirectional
Transmission power	20 mW

5.3. Evaluation parameters

In the performance evaluation, we measure the impact of increasing the EV density on system performance. The EV density is the number of EVs in the simulation scenario, and it is the most significant parameter because it is directly related to the traffic congestion. Every time we test the system under different EV density values and measure its effect on the system's performance, by increasing the EV density to higher values, we can test the scalability and robustness of the system under heavy EV traffic conditions.

5.4. Simulation environment

To perform the simulation, we have extracted the data of a real map from Beijing city from Open Street Map (OSM) [22]. As we needed a large area to test the performance with a large number of EVs and geographically distributed CS, we extracted a map of 5 km^2 using the GPS coordinates of the extracted area. The road information is taken from the raw data of the city map using OSM. However, the CS information was added later, as some of the areas have no CS at all. The generated EV traffic is populated using the Simulation of Urban Mobility (SUMO) platform [23], as shown in Fig. 4.

The proposed EV charging management scheme was simulated using OMNET++ network simulator [24] by defining the EV behaviors with the vehicular framework Veins [25]. Table 2 shows the details of the simulation parameters. Throughout the simulation time, three modules are running simultaneously: Veins are used for the network simulation, while SUMO runs the road traffic simulation and sumo-launched acts as an intermediate between the other two modules. Sumo frequently updates the traffic-related values using the Traci interface, as shown in Fig. 5.

The simulation details and the used tools, as well as the simulated map, are presented in Table 1. Whereas the detailed description of the EV wireless communication model used in the simulation is shown in Table 2.

The EVs join the network at a random intersection as the starting location and move to another randomly selected intersection as the ending location. Nonetheless, we modify the fringe factor to 10, which means that the edges of the simulated area located at the fringe of the city map are 10 times more likely to be selected as the start or end of a trip, which enables us to model a long-distance trip. Furthermore, the minimum driving distance is chosen as twice the average length of the roads, because there should be at least two consecutive roads for a meaningful trip, and the traffic lights are also selected as static, which means that each traffic light has a fixed duration regardless of the traffic conditions. That allows us to compute the expected arrival time in the road reservation operation. The EVs travel toward their destinations at a speed of

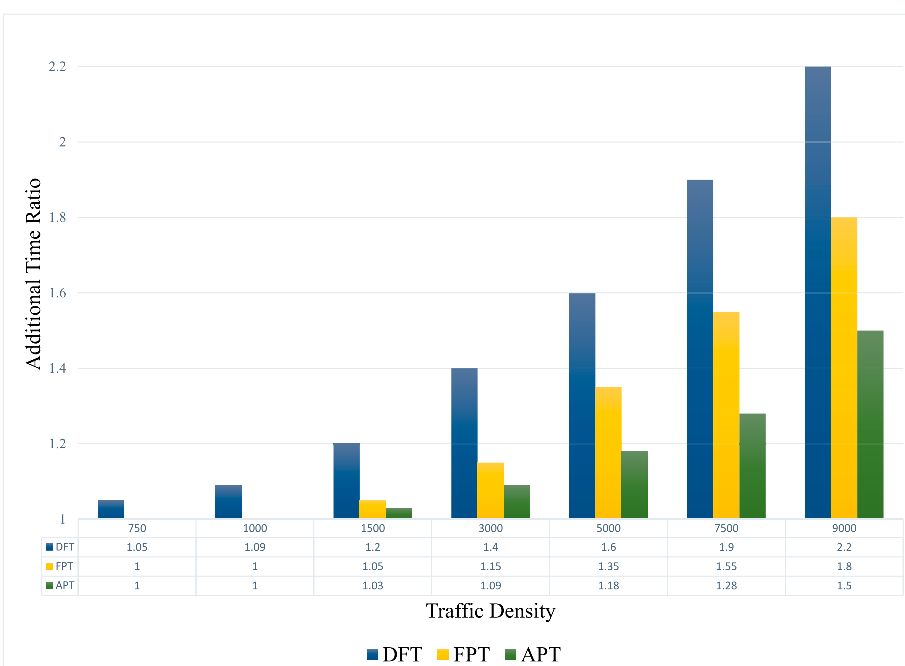

Fig. 6. Additional time vs. traffic density.

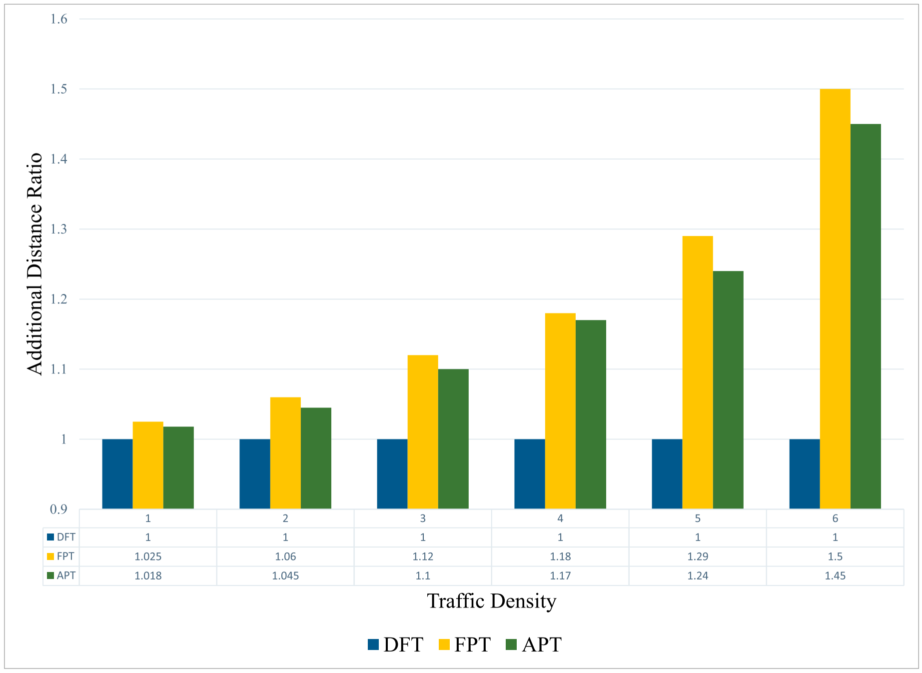

Fig. 7. Additional distance *vs.* traffic density.

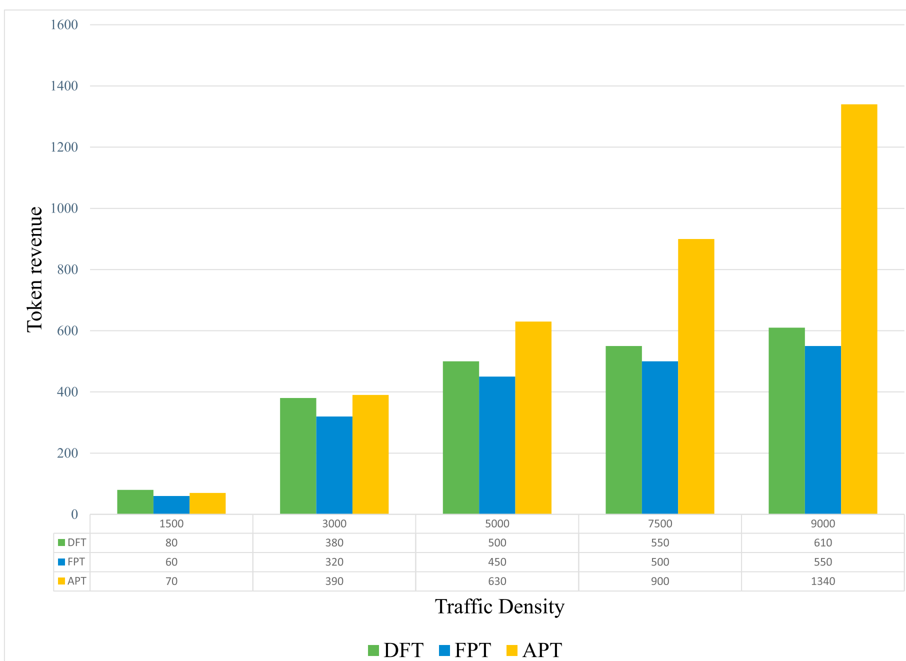

Fig. 8. Token revenue vs. traffic density.

(20–700) km/h. And all the presented results are the average of 100 simulation runs.

6. Results discussion

We tested the studied baselines (APT, DFT and FPT) and measured their TG, AD and AT, with increasing EV density settings. The results of EV density test are shown in Figs. 6–8. As we can notice in Fig. 6, in low EV density settings (under 450 EV), the AT of the studied baselines is nearly negligible. Nonetheless, when the EV density increases in (500–3000 EV) and (3000–9000 EV) settings, all of the three baselines' AT increase relative to the EV density, but to varying degrees. DFT's AT increases quickly with the increase of EV density, and reaches 2.3 at the highest density settings (9000). In other words, at 9000 EV density, the average travel time of EV to the destination is as high as 220% because EVs under DFT system always choose the shortest way even if it's congested. Unlike DFT, APT and FTP's AT moderately grow to score 1.45 at the highest EV density settings (9000), which is 50% less than FTP's because some EVs opt to go through the free-way instead of the paid congested way, which helps to mitigate the congestion and shorten the EV's travel time, hence decreasing the average AT.

In Fig. 7, the AD of the studied baselines in various EV density settings is shown. In low EV density settings (less than 700 EV), the AD of the studied baselines is nearly negligible, but in high and medium settings,

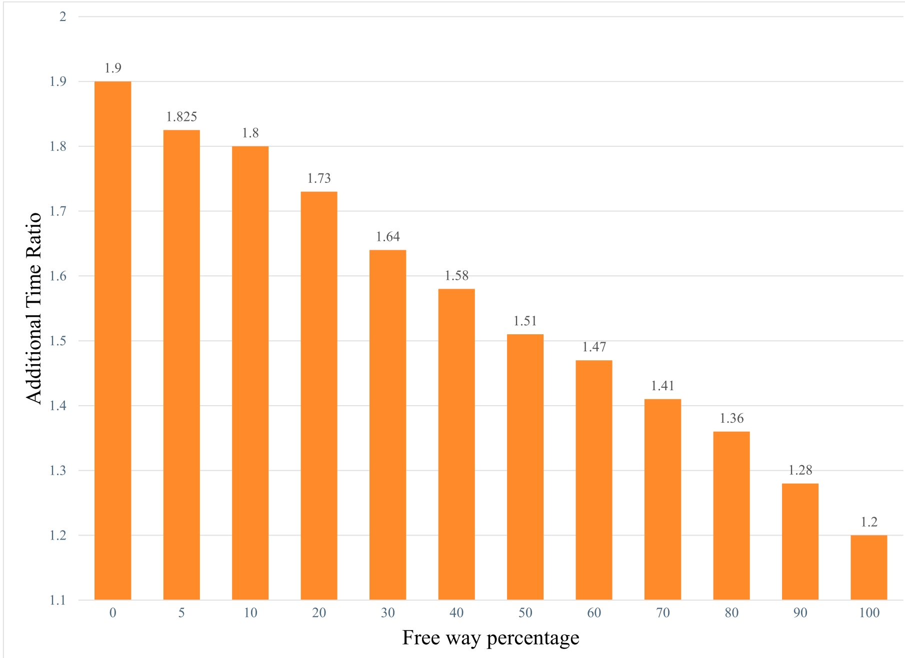

Fig. 9. The effects of free-way choice on average traveled time.

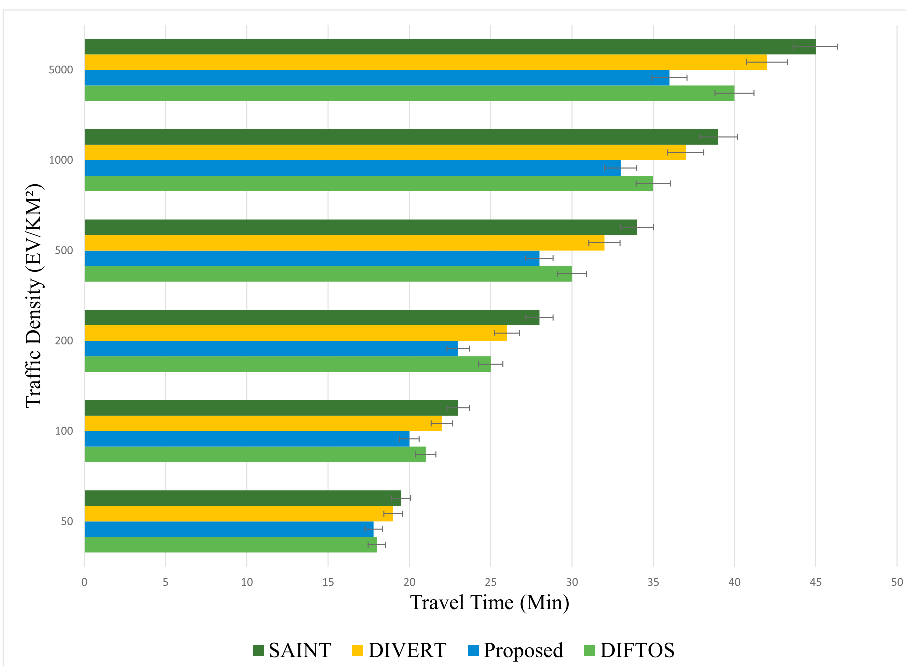

Fig. 10. Traffic density vs. average travel time.

FTP and APT's AD grow moderately to score 1.5 at the highest EV density because the alternative free-way are usually longer than the shortest way. Nevertheless, even at very high EV density settings, the vehicle will not travel an additional distance at DFT because the shortest road is always selected. Some may claim that DFT is preferable since it has no AD. It is true that the distance in DFT is better, and the EV does not have to go through additional distances to avoid the congestion. However, this comes at the cost of the huge increase in travel time, as we can see in Fig. 6. That is to say, in our proposed scheme, there is an increase in terms of additional distance because some of the EVs are willing to drive an additional distance and arrive ahead of time. For example, some may prefer to drive an extra 3 KM and arrive in 20 minutes, rather than 1.5 KM and arrive in 40 minutes and then pay tokens.

In Figs. 6 and 7, APT and FPT have almost the same scores. Nonetheless, Fig. 8 presents the superiority of our proposed system APT. In small EV density settings (up to 1100 EV), the token gain of the studied baselines is almost negligible because of the low EV traffic at the roads. Hence, the paid traffic quota is never used. However, in middle-traffic settings (2900–4900 EV), the token gains of the three baselines increase relatively to the EV density increase, but with different increase rates. Nonetheless, in high EV density settings (5000–9000 EV), although the token gain of DFT and FPT is stable at 500, it is due to their static paid

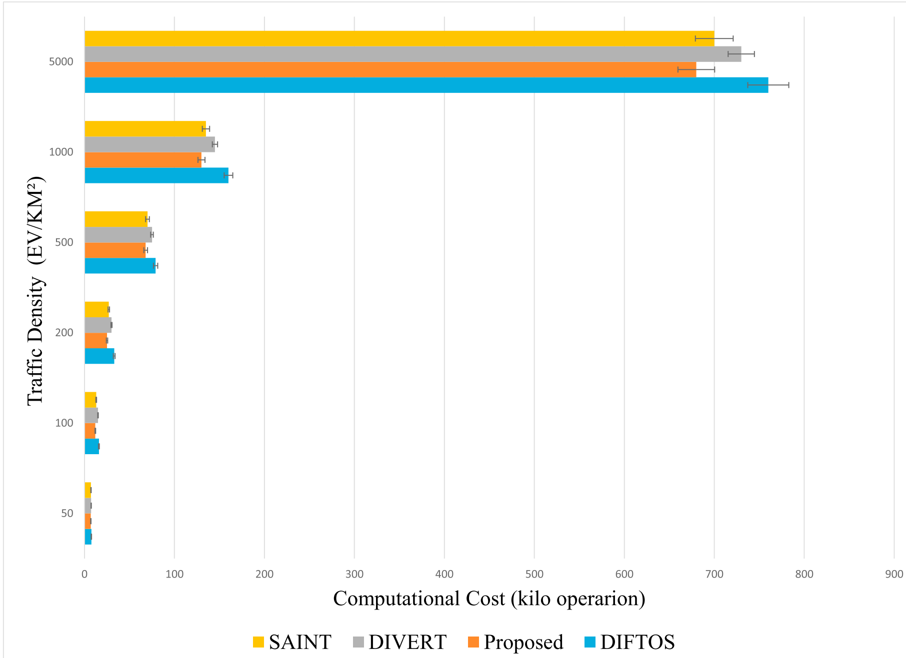

Fig. 11. Traffic density vs. traffic server computational cost.

charging quota. Nonetheless, APT gain grows quickly, resulting in its dynamic quota allocation and pricing. However, Fig. 9 presents the additional traveled time by changing the percentage of EVs' decisions to take the alternative free-way. From Fig. 9, we can see the negative correlation between the percentage of EVs' decisions to take alternative free-ways and the additional traveled time. This shows that the more EVs are willing to take the alternative way, the shorter the average traveled time.

In addition to the testing with the proposed system's variants as baselines, we have also tested it with the following congestion mitigation systems, SAINT [26], DIFTOS [27], and DIVERT [28]. We have tested the simulated baselines based on two testing metrics: the average travel time of all the simulated EVs, and the computational cost performed by the TEVC. Fig. 10 presents the average travel time of every baseline scheme with different traffic density parameters. Under different traffic settings, we can notice that the proposed scheme has the shortest travel time compared to other baselines, which is due to the proposed system alternative rerouting strategy. DIFTOS and DIVERT score better than SAINT, due to their distributed server architecture that enables the low-level servers to update and assign routes locally without asking the TEVC. Fig. 11 presents the computational cost in the TEVC for every system in different traffic conditions. Although the proposed system has the best performance when it comes to computing cost. DIFTOS has the worst performance. Nonetheless, it is worth noting that the computational resources in DIFTOS are distributed among various EVs' clusters that are distributed in the smart city, while SAINT and the proposed system TEVC are centralized, and DIVERT has a hybrid traffic server design.

7. Conclusions

In this work, we propose a novel dynamic traffic congestion pricing and EV charging management system for Internet of vehicles in the urban smart city environment. The proposed system gives incentive tokens to the EV that opts to take alternative congested free-ways and congested-free charging stations. We propose a token management system that serves as a kind of charging currency, where the vehicles earn these tokens if they take alternative non-congested ways and charging stations and use the tokens to pay for the charging fees. Charging tokens are also used in the tender operation between vehicles to manage the road reservation as well as charging reservation. The proposed scheme leverages dynamic pricing to meet the needs of traffic congestion in peak hours. Using extensive traffic simulation in different smart city scenarios, it is proved that the system can reduce the traffic congestion and the total charging time at the charging stations. Nonetheless, the proposed system could be improved in many aspects that we have left as future research directions:

1. In the current work, we believe that the static management of the traffic light, i.e., the combination of the dynamic traffic lights management system with the EV charging management system, is one of the future research directions.

2. In the proposed system, the server is considered as a centralized traffic center. Changing the server design to a distributed vehicular server (where the computational tasks of the server are performed by the EV) is also one of the future research directions.

Acknowledgements

This work is supported by the Fundamental Research Funds for Central Universities of China (No.FRF-GF-18-009B, No. FRF-BD-18-001A), the 111 Project (Grant No.B12012).

References

[1] D. Schrank, B. Eisele, T. Lomax, Tti's 2012 Urban Mobility Report, Texas A&M Transportation Institute. The Texas A&M University System vol. 4.
[2] K. Zhang, S. Batterman, Air pollution and health risks due to vehicle traffic, Sci. Total Environ. 450 (2013) 307–316.
[3] A. de Palma, R. Lindsey, Traffic congestion pricing methodologies and technologies, Transport. Res. C Emerg. Technol. 19 (6) (2011) 1377–1399.
[4] P. Hu, S. Dhelim, H. Ning, T. Qiu, Survey on fog computing: architecture, key technologies, applications and open issues, J. Netw. Comput. Appl. 98 (2017) 27–42, https://doi.org/10.1016/j.jnca.2017.09.002. URL, https://linkinghub.elsevier.com/retrieve/pii/S1084804517302953.
[5] N. Aung, W. Zhang, S. Dhelim, Y. Ai, Accident prediction system based on hidden markov model for vehicular ad-hoc network in urban environments, Information 9 (12) (2018) 311, https://doi.org/10.3390/info9120311. URL, http://www.mdpi.com/2078-2489/9/12/311.
[6] R.F. Atallah, C.M. Assi, W. Fawaz, M.H.K. Tushar, M.J. Khabbaz, Optimal supercharge scheduling of electric vehicles: centralized versus decentralized methods, IEEE Trans. Veh. Technol. 67 (9) (2018) 7896–7909. URL, https://ieeexplore.ieee.org/document/8369097/.
[7] R. Moghaddass, O.A. Mohammed, E. Skordilis, S. Asfour, Smart control of fleets of electric vehicles in smart and connected communities, IEEE Transactions on Smart

Grid 10 (6) (2019) 6883–6897. URL, https://ieeexplore.ieee.org/document/8700238/.
[8] M.E. Kabir, C. Assi, M.H.K. Tushar, J. Yan, Optimal scheduling of EV charging at a solar power-based charging station, IEEE Systems Journal 14 (3) (2020) 4221–4231. URL, https://ieeexplore.ieee.org/document/8998370/.
[9] R. Xie, W. Wei, Q. Wu, T. Ding, S. Mei, Optimal service pricing and charging scheduling of an electric vehicle sharing system, IEEE Trans. Veh. Technol. 69 (1) (2020) 78–89. URL, https://ieeexplore.ieee.org/document/8887240/.
[10] N.T. Milas, D.A. Mourtzis, P.I. Giotakos, E.C. Tatakis, Two-layer genetic algorithm for the charge scheduling of electric vehicles, in: 2020 22nd European Conference on Power Electronics and Applications (EPE'20 ECCE Europe), IEEE, 2020. P.1–P.10, https://ieeexplore.ieee.org/document/9215685/.
[11] Z. Liu, Q. Wu, S.S. Oren, S. Huang, R. Li, L. Cheng, Distribution locational marginal pricing for optimal electric vehicle charging through chance constrained mixed-integer programming, IEEE Transactions on Smart Grid 9 (2) (2018) 644–654. URL, http://ieeexplore.ieee.org/document/7464322/.
[12] E.S. Rigas, S.D. Ramchurn, N. Bassiliades, G. Koutitas, Congestion management for urban EV charging systems, in: 2013 IEEE International Conference on Smart Grid Communications (SmartGridComm), IEEE, 2013, pp. 121–126. URL, http://ieeexplore.ieee.org/document/6687944/.
[13] Y. Huang, K.M. Kockelman, Electric vehicle charging station locations: elastic demand, station congestion, and network equilibrium, Transport. Res. Transport Environ. 78 (2020) 102179. URL, https://linkinghub.elsevier.com/retrieve/pii/S1361920919306406.
[14] X. Luo, R. Qiu, Electric vehicle charging station location towards sustainable cities, Int. J. Environ. Res. Publ. Health 17 (8) (2020) 2785. URL, https://www.mdpi.com/1660-4601/17/8/2785.
[15] Y. Li, H. Celebi, M. Daneshmand, C. Wang, W. Zhao, Energy-efficient femtocell networks: challenges and opportunities, IEEE Wireless Communications 20 (6) (2013) 99–105. URL, http://ieeexplore.ieee.org/document/6704480/.
[16] Y. Li, C. Liao, Y. Wang, C. Wang, Energy-efficient optimal relay selection in cooperative cellular networks based on double auction, IEEE Trans. Wireless Commun. 14 (8) (2015) 4093–4104. URL, http://ieeexplore.ieee.org/document/7069238/.
[17] Y. Li, J. Liu, B. Cao, C. Wang, Joint optimization of radio and virtual machine resources with uncertain user demands in mobile cloud computing, IEEE Trans. Multimed. 20 (9) (2018) 2427–2438. URL, https://ieeexplore.ieee.org/document/8265188/.
[18] N. Aung, W. Zhang, S. Dhelim, Y. Ai, T-Coin, Dynamic traffic congestion pricing system for the internet of vehicles in smart cities, Information 11 (3) (2020) 149. URL, https://www.mdpi.com/2078-2489/11/3/149.
[19] Y. Zhu, Z. Li, H. Zhu, M. Li, Q. Zhang, A compressive sensing approach to urban traffic estimation with probe vehicles, IEEE Trans. Mobile Comput. 12 (11) (2012) 2289–2302.
[20] S. Dhelim, N. Huansheng, S. Cui, M. Jianhua, R. Huang, K.I.-K. Wang, Cyberentity and its consistency in the cyber-physical-social-thinking hyperspace, Comput. Electr. Eng. 81 (2020) 106506. URL, https://linkinghub.elsevier.com/retrieve/pii/S0045790618334839.
[21] X. Cai, H. Ning, S. Dhelim, R. Zhou, T. Zhang, Y. Xu, Y. Wan, Robot and its living space A roadmap for robot development based on the view of living space, Digital Communications and Networks doi:10.1016/j.dcan.2020.12.001. URL https://linkinghub.elsevier.com/retrieve/pii/S2352864820302881.
[22] M. Haklay, P. Weber, OpenStreetMap: user-generated street maps, IEEE Pervasive Computing 7 (4) (2008) 12–18. URL, http://ieeexplore.ieee.org/document/4653466/.
[23] M. Behrisch, L. Bieker, J. Erdmann, D. Krajzewicz, SUMO–simulation of urban mobility: an overview, in: Proceedings of SIMUL 2011, the Third International Conference on Advances in System Simulation, ThinkMind, 2011.
[24] A. Varga, R. Hornig, An overview of the OMNeT++ simulation environment, in: Proceedings of the 1st International Conference on Simulation Tools and Techniques for Communications, Networks and Systems & Workshops, ICST (Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering), 2008, p. 60.
[25] H. Noori, Realistic urban traffic simulation as vehicular Ad-hoc network (VANET) via Veins framework, in: 2012 12th Conference of Open Innovations Association (FRUCT), IEEE, 2012, pp. 1–7. URL, http://ieeexplore.ieee.org/document/8122092/.
[26] J. Jeong, H. Jeong, S. Member, E. Lee, T. Oh, S. Member, D.H.C. Du, Saint : self-adaptive interactive navigation tool for cloud-based vehicular traffic optimization, IEEE Trans. Veh. Technol. 65 (6) (2016) 4053–4067.
[27] W. Zhang, N. Aung, S. Dhelim, Y. Ai, DIFTOS: a distributed infrastructure-free traffic optimization system based on vehicular ad hoc networks for urban environments, Sensors 18 (8) (2018) 2567. URL, http://www.mdpi.com/1424-8220/18/8/2567.
[28] J.S. Pan, I.S. Popa, C. Borcea, Divert : a distributed vehicular traffic Re-routing system for congestion avoidance, IEEE Trans. Mobile Comput. 16 (1) (2017) 58–72.